Charan Deep Danthika

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Professional Summary

Computer Science undergraduate with a strong foundation in **object-oriented design, data structures, algorithms, and complexity analysis**, and hands-on **Python** development experience. Architected a full-stack, **scalable RAG platform** with hybrid retrieval, a pluggable vector-store layer, and multi-provider service routing, alongside ML pipelines processing 27,000+ records. Comfortable designing solutions to broadly-defined, ambiguous problems in agile environments; seeking a **Software Development Engineer Intern** role to contribute to large-scale distributed systems.

Education

Rajeev Gandhi Memorial College of Engineering and Technology **India**
Bachelor of Technology in Computer Science and Engineering *Year of Graduation: 2027*

- **CGPA:** 8.02 / 10.0
- **Coursework:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Machine Learning

Technical Skills

Cloud, Linux & DevOps: AWS (EC2, S3, RDS, IAM, VPC), Linux CLI (Bash), Docker, Docker Compose
Computer Science Fundamentals: Data Structures & Algorithms, Object-Oriented Design, Complexity Analysis, DBMS
Languages & Data Tools: C, C++, Java, Python, SQL, JavaScript
Generative AI & NLP: RAG pipeline development, Sentence-Transformers, FAISS (vector search), Prompt Engineering, LLM API integration (OpenAI, Gemini), Agentic tool-calling
Machine Learning: Exploratory Data Analysis, Classification Models (XGBoost, CatBoost, Random Forest), Hyperparameter Tuning, Model Evaluation, Matplotlib/Seaborn
Web & Backend: React.js, Node.js, FastAPI, REST API design & integration, HTML/CSS
Databases: MySQL, SQLite
Tools & Concepts: Git/GitHub, VS Code, Jupyter Notebook, OOP
Soft Skills: Problem-Solving, Communication, Team Collaboration, Adaptability

Experience

Cloud Computing Intern – AWS Services **Remote**
EduSkills Foundation (in association with AWS Academy) *2024*

- Completed a structured internship on AWS services: provisioned and configured EC2 instances, S3 buckets, RDS databases, IAM roles, and VPC networking.
- Gained hands-on experience with cloud architecture design, cost optimization, and secure, scalable deployment on AWS.

Projects

Resume RAG Assistant | *FastAPI, Next.js, FAISS/ChromaDB, JWT, Docker, Groq/OpenAI/Gemini* | GitHub

- Architected a **scalable**, full-stack RAG platform, containerized with **Docker and Docker Compose**, with a **hybrid retrieval pipeline** combining BM25 keyword search and dense vector embeddings, fused via **Reciprocal Rank Fusion** and cross-encoder reranking.
- Built a multi-provider **LLM router** (OpenAI, Gemini, Groq, local Ollama) with a **ReAct-style tool-calling agent** that streams grounded, cited responses to a Next.js frontend via Server-Sent Events.
- Designed a pluggable vector-store layer (FAISS/ChromaDB) with JWT-authenticated document upload, magic-byte file validation, and an in-memory query cache for low-latency repeat searches.

College Enquiry Chatbot | *Python, SQL, REST APIs, Web* | GitHub

- Developed a chatbot to automate student inquiries regarding courses, admissions, and faculty information, resulting in a 60% reduction in manual front-desk workload.
- Integrated SQL-based query handling with backend logic to serve dynamic, real-time responses.
- Built a responsive web interface for student interaction; reduced repetitive administrative queries by **60%** with **85%** response accuracy across internal testing.

Student Depression Prediction | *Python, Scikit-learn, XGBoost, CatBoost* | GitHub

- Built an end-to-end ML pipeline on a **27,900+** record student dataset to predict depression risk from academic, financial, and lifestyle factors.
- Performed EDA and feature engineering across 18 features (academic pressure, sleep duration, financial stress, family history, etc.); trained and compared 7 classification models.
- Tuned hyperparameters on ensemble models; best-performing Gradient Boosting model achieved **84.8%** accuracy and **0.918** ROC-AUC on held-out data.

Certifications

Oracle Cloud Infrastructure 2025 Certified Generative AI Professional | *Oracle University*

Microsoft Certified: Azure Data Fundamentals | *Microsoft Learn*

Supervised Machine Learning: Regression and Classification | *DeepLearning.AI & Stanford University (Andrew Ng)*

Leadership

- **DSA Club** — organized coding workshops and peer learning sessions; mentored juniors on problem-solving and coding best practices; coordinated competitions and designed practice challenges to build community engagement.